

3.1 Programming

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 01 of 24

3.1 Programming

LEARN IT

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own. They need exact, step-by-step **instructions**.

- A set of instructions given to a computer is called a **program**.
- The language used to write these instructions is called a **programming language**.
- Writing programs is called **programming**, and the person who writes them is a **programmer**.

Think of a recipe. A cook follows the recipe line by line: "boil water, add rice, wait 15 minutes". A computer follows a program in the same way, one instruction at a time, without guessing what you meant.

WORKED EXAMPLE

A tiny program. Here is a complete Python program with one instruction. It tells the computer to show a message on the screen.

```
print("Hello")
```

OUTPUT
Hello

The word `print` is the instruction; `"Hello"` is the data it works on.

REMEMBER IT

Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix it for you.

COMMON MISTAKE

Students often think the computer "understands" what they want. It does not. If you type `Print("Hello")` with a capital P, Python stops with an error, because it only knows `print`. Be exact.

اردو میں سمجھیں

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔ اسے کام کرنے کے لیے ہدایات چاہئیں۔ ہدایات کے مجموعے کو پروگرام (Program) کہتے ہیں، اور جس زبان میں یہ ہدایات لکھی جاتی ہیں اسے پروگرامنگ لینگویج (Programming Language) کہتے ہیں۔

CHECK YOURSELF

1. What is a program?
2. Why do we say computers are "dumb" machines?

Answers: 1. A set of instructions given to a computer to perform a task. 2. Because they cannot think or do anything without instructions; they only follow exactly what they are told.

Next lecture: 3.1.1 Programming Language Types